Water Jug Problem

Aim

To solve the **Water Jug Problem** using Python by finding the sequence of steps to obtain the required amount of water.

Algorithm

1. Initialize two jugs with capacities m and n.
2. Fill one of the jugs.
3. Pour water from one jug to the other until one becomes empty or the other becomes full.
4. Empty a jug whenever necessary.
5. Repeat the process until the desired quantity is obtained.

Python Code

```
from collections import deque


def water_jug(jug1, jug2, target):
    visited = set()
    queue = deque([(0, 0)])

    while queue:
        x, y = queue.popleft()

        if (x, y) in visited:
            continue
        visited.add((x, y))

        print((x, y))

        if x == target or y == target:
            print("\nTarget reached!")
```

```
    return

  next_states = [
    (jug1, y),            # Fill Jug1
    (x, jug2),            # Fill Jug2
    (0, y),               # Empty Jug1
    (x, 0),               # Empty Jug2
    (max(0, x-(jug2-y)), min(jug2, y+x)),  # Pour Jug1 -> Jug2
    (min(jug1, x+y), max(0, y-(jug1-x)))   # Pour Jug2 -> Jug1
  ]

  for state in next_states:
    if state not in visited:
      queue.append(state)

water_jug(4, 3, 2)
```

Output

```
.py
(0, 0)
(4, 0)
(0, 3)
(4, 3)
(1, 3)
(3, 0)
(1, 0)
(3, 3)
(0, 1)
(4, 2)

Target reached!
```

Result

The **Water Jug Problem** was successfully solved using Python. The program generated the sequence of valid states and reached the required target quantity of water successfully.